PROJECT DEFINITION / 12 NOVEMBER 2026

Roofscapes (Re)Powered

AI-assisted solar investment feasibility and collective self-consumption for commercial building owners

THE CORE DECISION
Should a commercial property owner install PV across the feasible rooftop area, and can an eligible group of nearby consumers make that investment more valuable than a host-only arrangement?

One clearly defined user

Primary user: commercial property owner, asset manager, or authorized institutional property manager assessing a rooftop PV retrofit. Secondary stakeholders: solar developer, energy consultant, participating residents/businesses, and a collective self-consumption manager.

The project in one sentence

VALUE PROPOSITION
Identify feasible rooftop PV capacity, predict when generation exceeds the host building's demand, find nearby consumers with complementary hourly load, and compare legally constrained sharing agreements against a host-only investment.

Why it fits the AEC / Smart Construction studio

- Design and Pre-Construction: supports PV retrofit sizing, feasibility, economic assessment and stakeholder decisions before installation.
- Operation: uses demand profiles to simulate how the installed system would perform once operating.
- Verified AI: tests a demand-prediction model against rule-based schedules and held-out metered records; optimization remains transparent and constraint-based.

Research hypothesis: Realistic hourly demand prediction plus constrained collective allocation can improve estimated economic value and PV self-consumption relative to simple host-only and fixed-profile baselines, for some building/community combinations.

Scope: build one credible investment decision, not an energy marketplace

IN SCOPE - DECEMBER PROOF	OUT OF SCOPE - DO NOT BUILD
One host office, school, retail or similar commercial rooftop in the existing nine-block study area.	Automated citywide community formation or permanent real-time energy trading.
3-5 nearby candidate buildings with stated eligibility assumptions.	Signing contracts, setting tariffs, live electricity billing or guaranteed savings.
Hourly PV generation, hourly demand, fixed and pre-agreed variable allocation scenarios.	Storage, grid dispatch, physical peer-to-peer wiring or smart-meter control.
Host-only vs collective scenarios and annual cost/benefit comparison with uncertainty.	Structural sign-off, permit approval or legally certified eligibility.

Target product experience

USER ACTION	SYSTEM RESPONSE
1. Select a commercial host building.	Building ID, roof geometry, preliminary installable kWp, data provenance.
2. Enter or upload the host hourly load.	Weekday / weekend / seasonal demand curves and solar-hour mismatches.
3. Inspect eligible nearby buildings.	Candidate list with distance, use type, assumed demand, and confidence.
4. Compare allocation strategies.	Host-only, fixed coefficients, pre-defined hourly-variable coefficients.
5. Review recommendation.	Estimated self-consumed kWh, residual export, owner / participant economics, payback and caveats.

IMPORTANT REGULATORY FRAMING
Collective self-consumption allocates metered generation through previously agreed coefficients. Hourly-variable coefficients may be specified in advance; the tool must not promise spontaneous, retrospective or physically directed transfers to chosen flats. Eligibility depends on applicable legal, connection and metering conditions.

Step-by-step development pipeline: inputs to outputs

The shaded boxes are the smallest complete data flow for a demonstrable final prototype. Keep measured, modeled and assumed values clearly labeled.

01 Select host	INPUT: Address / parcel ID; building use; roof / ownership information	OUTPUT: Resolved building footprint, coordinates, roof ID, cadastral attributes
02 Evaluate PV capacity	INPUT: ICGC LiDAR + imagery; obstacle masks; panel specification; exclusion rules	OUTPUT: Usable m2, panel count, installed kWp, layout and confidence
03 Simulate generation	INPUT: Panel layout; LiDAR shadow horizon; PVGIS hourly weather / radiation	OUTPUT: Timestamped hourly PV generation G[h], annual kWh, shading losses
04 Read host demand	INPUT: Authorized 15/30/60-minute meter data OR proxy profiles; use; floor area; weather	OUTPUT: Hourly demand D[host,h]; calendar features; quality flags
05 Forecast demand (AI)	INPUT: Historical load; weekday/weekend; temperature; month; building category	OUTPUT: Predicted hourly load and error metrics with confidence band
06 Discover neighbours	INPUT: Candidate coordinates; building types; electrical/network eligibility evidence	OUTPUT: Shortlist of 3-5 candidates and eligibility flags
07 Optimize allocation	INPUT: G[h]; all D[i,h]; tariff scenarios; coefficient and contract constraints	OUTPUT: Fixed / predefined variable beta[i,h]; allocated kWh; export
08 Compare investment	INPUT: Installed costs; O&M; retail tariffs; compensation; discount rate	OUTPUT: Incremental benefit, self-consumption ratio, payback and scenario sensitivities
09 Publish report	INPUT: All previous results; sources; assumptions; validation results	OUTPUT: 3D host/community view + exportable feasibility report

Core modules: spatial lookup -> roof/PV simulation -> demand ingestion -> ML demand forecast -> eligibility screening -> allocation optimizer -> financial evaluation -> UI/report.

Data sources, access and reliability

SOURCE	FORMAT	USES	ACCESS / LIMITATION
ICGC LiDAR / orthophoto	Point cloud / raster or WMS	Roof shape, heights, nearby shading and obstacles	Public; verify acquisition year, coverage and licensing
Catastro / ICGC geocoder	Web service / GML / structured data	Building footprint, address, use, cadastral attributes	Public; property owner / tenancy not established
European Commission PVGIS	API JSON / CSV; hourly / TMY	Reference irradiation, weather and predicted PV power	Public; modeled estimate, not meter truth
Barcelona Open Data electricity consumption	CSV; postcode, sector, time band	Consumption-pattern proxy / sector-level context	Public; aggregated, not a building hourly meter
Datadis / owner-authorized utility readings	Authorized interval meter exports	Host and participant actual hourly load; AI target labels	Permission dependent; privacy and coverage risks
Building owner documents / installer offers	Bills, floor area, roof documents, quotations	Validated inputs, cost/ownership assumptions	Must be supplied/approved by the owner
IDAE / MITECO rules	Guides + current regulation	Sharing coefficients, modalities, legal checks	Use as validation constraints; verify case by case
Weather observations (Meteocat / ICAEN)	Station tables and time series	Weather comparison and predictive features	Station is not roof-specific irradiance

CRITICAL DATA GAP
Barcelona's open electricity dataset provides consumption aggregated by postal code, economic sector and time band. It cannot establish an individual office or apartment's true hourly demand. To claim validated building-level demand forecasting, obtain authorized interval meter data or a trustworthy open benchmark and clearly identify transfer limitations.

Four-week delivery plan

Four development sprints from 12 November to 10 December 2026, followed by final polishing before the 17 December defense. Each sprint ends with a testable artifact.

WEEK 1 / NOV 12-18

DATA + SPECIFICATION

- Freeze host type, nine-block geography, sharing modality and one success metric.
- Choose one real host building; retrieve existing PV hourly series and roof confidence.
- Acquire owner-authorized hourly load or select a clearly labeled proxy fallback.
- Define schema, dataset provenance, quality checks and SDD project constitution.
- Connect one live verified source through the prescribed MCP/connector boundary.

Exit test: Gate: host PV + host load aligned into one 8,760-hour table; source metadata recorded.

WEEK 2 / NOV 19-25

DEMAND + AI BASELINE

- Create deterministic building-type weekday/weekend/seasonal profiles.
- Engineer demand features from meter records: hour, weekday, holiday, temperature.
- Train a small gradient-boosting model only if usable labeled records exist.
- Compare MAE, RMSE, bias vs schedule baseline on held-out dates/buildings.
- Map 3-5 nearby candidates and mark data provenance/eligibility as tentative.

Exit test: Gate: reproducible demand model comparison, or documented no-ML fallback with reason.

WEEK 3 / NOV 26-DEC 2

SHARING + ECONOMICS

- Build hourly energy accounting for host-only, fixed beta and pre-agreed hourly beta.
- Implement optimizer with beta sum-to-one, non-negative and eligibility constraints.
- Estimate household/host allocations, consumption, export and financial effects.
- Test weekday/weekend/seasonal scenarios and sensitivity to load uncertainty.
- Connect pipeline results to 3D community map and a simple scenario comparison.

Exit test: Gate: transparent simulation with at least two baselines and one optimization scenario.

WEEK 4 / DEC 3-10

VALIDATE + DEPLOY

- Unit test hourly energy conservation, coefficient validity and time alignment.
- Run sensitivity analysis for tariffs, generation, occupancy and cost assumptions.
- Benchmark demand where independent meter data exists; obtain expert feedback.
- Prepare report and UI with flags for modeled versus observed inputs.
- Finalize reproducible demonstration, blog draft, pitch and limitations.

Exit test: Gate: end-to-end demonstration, tests, documented failures and baseline comparison.

Input / output specifications for implementation

Canonical timestamps: Europe/Madrid with UTC storage; explicitly handle daylight-saving-time transitions. All energy flow is represented in hourly kWh.

FILE / DATA OBJECT	REQUIRED FIELDS	CONSUMER / PURPOSE
host_building.json	cadastral_id, lat, lon, use_type, roof_area_m2, valid_date	Single selected host identity and audit information
pv_hourly.csv	timestamp_utc, generation_kwh, irradiance_source, shade_loss, model_version	Host hourly solar generation G[h]
demand_hourly.csv	building_id, timestamp_utc, consumed_kwh, source_type, quality_flag	Observed or modeled load D[i,h]
candidate_buildings.geojson	building_id, use_type, geometry, distance_m, eligible_status	Spatial list of potential participants
allocation.csv	timestamp_utc, building_id, beta, allocated_kwh, used_kwh, excess_kwh	Hourly accounting under each scenario
investment_summary.json	scenario, capex_eur, annual_savings_eur, export_value_eur, simple_payback_y	Comparable techno-economic outputs
validation_report.json	baseline_name, metric, value, sample_n, split_method, failures	Traceable measured performance
report.pdf / web view	all calculated files + explanations + provenance	User-facing decision support, never automatic approval

Simplified energy allocation mathematics

For each hour h: G[h] = PV generation; D[i,h] = consumer i demand; beta[i,h] = agreed energy-allocation coefficient.

CHECKS
For each hour, beta[i,h] >= 0 and the sum of beta[i,h] across eligible participants = 1. Assigned energy A[i,h] = beta[i,h] x G[h]. Self-consumed energy S[i,h] = min(A[i,h], D[i,h]). Unused allocation and network export are handled per the applicable billing modality. Coefficients must be pre-agreed rather than retrospectively reassigned.

Minimum comparison metrics

- Energy: annual generated kWh, aggregate self-consumed kWh, fraction exported and annual site self-consumption ratio.
- Economic: annual net value (EUR), owner/participant benefit under stated tariff assumptions, simple payback and incremental gain vs host-only.
- AI: demand MAE/RMSE on held-out observations; comparison against fixed schedules.
- Operability: run time for the scenario, reproducibility and successful end-to-end demo.

Validation, guardrails and decision points

RISK / LIMIT	MITIGATION / REQUIRED TEST
No building-level metered demand	Use transparent representative demand assumptions; label all savings as scenarios. Do not claim trained demand accuracy without ground truth.
Model overfits one property / season	Time-based test split; ideally hold out a building; report accuracy by hour, weekday and season.
Invalid legal or electrical eligibility	Candidate screening only. Require a professional check of proximity, connection, metering, modality and signed allocation agreement.
Incorrect seasonal/time matching	Hourly UTC alignment, local display, DST unit tests, no transferring daytime generation to nighttime without storage.
Unverified PV or construction feasibility	Keep engineering, heritage, access and permit flags; no automated building-safety claims.
Finances too optimistic	Use multiple tariff / capex scenarios; show export value and operating costs; avoid guaranteed payback claims.
Privacy / data permission	Store no unnecessary personal information; require access authorization for interval meter records.

Development decisions to resolve in the first sprint

- Host: Which real commercial roof and owner/manager can provide an hourly demand profile?
- Participants: Which 3-5 candidate buildings have accessible consumption evidence?
- Optimization objective: Maximize collective self-consumption, owner net value, total participant savings, or a weighted objective?
- Business model: Who pays the PV capex, who pays the roof owner, and how are bill savings shared?
- Regulatory scope: Which collective self-consumption modality and connection regime will be used for the demo?

DECEMBER 17 DEMONSTRATION
Select one host -> inspect roof and projected PV output -> load demand -> identify eligible neighbours -> compare host-only and collective allocation -> review the forecast savings, validation results and the assumptions that still require professional verification.

Evidence base and source links

IAAC MaAI01 Research Studio 2026-27 syllabus
User-provided studio document. Defines AEC phases, SDD/harness, MCP, baseline comparison and final demonstration.

Roofscapes (Re)Powered - November 2026 midterm deck
User-provided pitch; preliminary nine-block study, rooftop/PVGIS pipeline and sharing demonstration.

Roofscapes project GitHub
https://github.com/iaac-maai/research_studio_team_1

IDAE - Guia de autoconsumo colectivo
https://www.idae.es/publicaciones/guia-de-autoconsumo-colectivo

MITECO - Preguntas frecuentes de autoconsumo
https://www.miteco.gob.es/es/energia/energia-electrica/electricidad/autoconsumo-electrico/preguntas-frecuentes-autoconsumo.html

European Commission JRC - PVGIS API
https://joint-research-centre.ec.europa.eu/photovoltaic-geographical-information-system-pvgis/using-pvgis-5/api-non-interactive-service_en

Barcelona Open Data - electricity by postcode, sector and time band
https://opendata-ajuntament.barcelona.cat/data/en/dataset/consum-electricitat-bcn

ICGC Territorial Orthophoto
https://www.icgc.cat/en/Geoinformation-and-Maps/Data-and-products/Image/Territorial-Orthophoto

ICGC Territorial LiDAR
https://www.icgc.cat/en/Geoinformation-and-Maps/Data-and-products/Elevations/Territorial-elevations/Territorial-LiDAR

ICAEN Solar Radiation Atlas
https://atlesradiaciosolar.icaen.gencat.cat/

EVIDENCE STATUS
This document is a proposed implementation plan, not evidence that the revised demand model, optimization, tariff analysis or legal screening has already been validated. Existing roof/PV and web components are described from the team's shared research materials; their live status needs confirmation against local development branches.